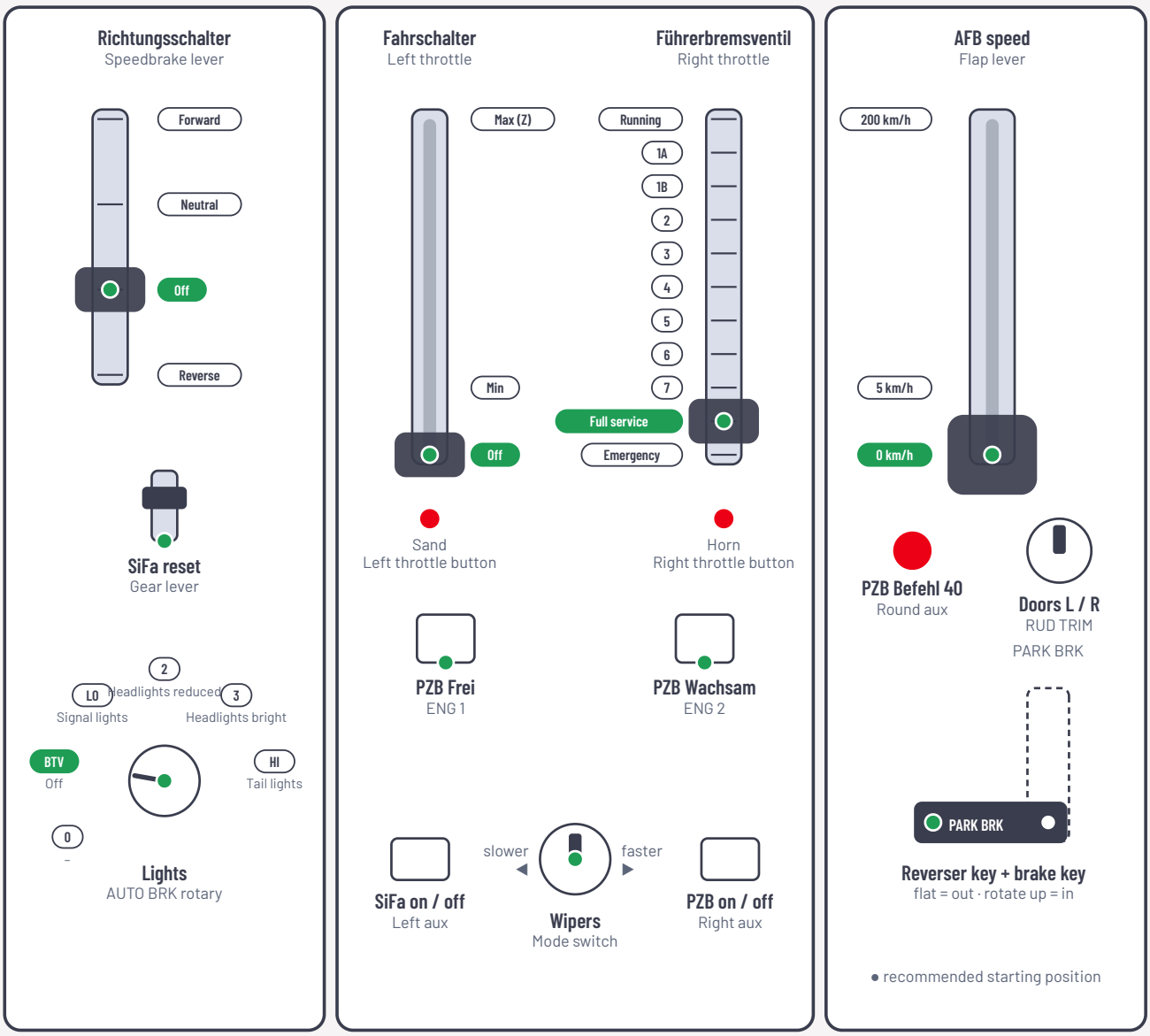
BR 101 Expert



Cab setup

- Machine room first: cut-off valves open, LSS 44F01 set (page 2)
- Fahrschalter at Off; PARK BRK up: reverser key in, brake key unlocked (held while up)
- Richtungsschalter to Neutral, wait for the screens; pantograph up, Hauptschalter in
- Brake preparation, SiFa and PZB on (aux squares), tests (page 2)
- Train data on the ZDE panel; lights to Signal lights or Headlights
- Forward, ZS train supply on, doors (RUD TRIM); AFB on in the cab

Cleared to depart

- Close doors, check the signal and PZB
- Parking brake off on the back wall (needs 1.5 bar C-pressure)
- Führerbremsventil to Running, release the direct brake
- Fahrschalter towards Z, or set the AFB speed on the flap lever
- Acknowledge SiFa with the gear lever, PZB with the ENG switches
- Brake: Führerbremsventil towards SB in steps



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BR 101 Expert Aufrüsten: from cold to departure

A cold-start service finds the BR 101 dark with the parking brake on; a timetable service loads it with the cab unoccupied, SiFa isolated and the LZB/PZB breaker open. On a loco that is already powered, start at step 8.

Back wall and machine room

1. **CAB 1 BACK WALL** Loco under the wire; parking brake light (Federspeicher anlegen) on.
2. **CAB 1 BACK WALL** Battery (Batterie) ON; the voltage must read above 95 V.
3. **CAB 1 BACK WALL** Pantograph selector to Auto (or the one you want); every LSS breaker on this wall set.
4. **VALVE CABINET** Open the SIFA and the red PZB/LZB cut-off valves; all others open except drain, towing, bypass and relay.
5. **BREAKERS BY CAB 2** Set every breaker, including LSS 44F01 LZB/PZB-Hauptschalter in the third row.
6. **MACHINE ROOM** Auxiliary reservoir under 7 bar: close LSS 34F05 (auxiliary compressor); if it will not run, drain below 4.8 bar first.
7. Leave the machine room before raising the pantograph: entering it with the Hauptschalter closed is prohibited.

Occupy Cab 1 and power up

8. Sit down; Fahrschalter (power handle) at Off, or the reverser will not move. (Left throttle)
9. Reverser key in and brake key unlocked; keep PARK BRK up, the key is held only while it is. (PARK BRK)
10. Richtungsschalter from Off to Neutral; wait for the screens to boot (over a minute). (Speedbrake lever)
11. **DESK** Raise the pantograph (SA heben), close the Hauptschalter (HS ein). 'Druckluft HS zu gering' = under 7 bar.
12. **LEFT SCREEN** Press [St] and check for faults.
13. Lights as required; off on an end coupled to a coach. (AUTO BRK)

Brake system preparation

14. **BACK WALL** NBÜ/ep switch to 0 in both cabs; pn/el switch to pn.
15. Führerbremsventil to Running; ease the direct brake until the C-pressure is below 0.5 bar. (Right throttle)
16. Parking brake light stays lit; wait for the main reservoir (red needle) to reach 10 bar.
17. pn test: brake pipe down to 4.2 bar, C-pressure rises on both bogies; release. (Right throttle)
18. pn/el switch to el and repeat the 4.2 bar application and release. (Right throttle)
19. **DESK** Direct brake to 2 bar C-pressure, release, then apply it again to hold the loco.

Safety systems and tests

20. SiFa and PZB to normal (LZB / INDUSI buttons on the ZDE panel). LZB stays isolated only light or under four braked coaches. (Left aux) (Right aux)
21. For the PZB/LZB test: Forward, brake key unlocked, train brake released, direct brake on. (Speedbrake lever)
22. **MFA** Press Prüftaste/Stör: lamps and sounds, two emergency applications, LZB calibration; then Prüftaste Lampen.
23. **SIFA PEDAL** SiFa test: hold the pedal (Q): light at 30 s, 'SIFA' 2.5 s later, Zwangsbremsung 2.5 s after that.
24. Test the horn. (Right throttle button)
25. **BRAKE KEY IN THE CAB** Brake test: brake pipe 5 bar, brake key locked 60 s, it may drop 0.3 bar at most; unlock.
26. Brake to 4.2 bar; with a cab car lock the brake key 10 s, the pipe must not rise. (Right throttle)
27. **OUTSIDE** Walk the train: every coach braked, then released. A sticking coach: overcharge to 5.3–5.5 bar.
28. **CAB CAR** With a cab car: hold Continuity Test until the main reservoir drops below 5 bar; it must climb to 8.

Train data (ZDE)

29. **ZDE PANEL** BRA: 9 for passenger; 2 freight in G, 4 freight in P.
30. **EBULA** Brh: EBuLa [0], [9] SD, [2] Brh berechnen; enter braked weight and mass, [E].
31. **BACK WALL SHEET** Convert it with the ZDE sheet (Brh im Zug) to the BRH to enter; a typical IC is 180, entered as 18.
32. ZL: length in tens of metres, rounded up (254 m = 26). VMZ: lowest limit in tens (200 km/h = 20).
33. **ZDE PANEL** Set BRA, BRH, ZL, VMZ, press ZD. [55] and [Stör] flashing = base data (Grunddaten) still active.
34. Check: hold PZB Wachsam and PZB Frei together; the MFA shows the data. (ENG 1 + ENG 2)
35. **LEFT SCREEN** Enter the train number and set up the GSM-R radio.

Departure, and leaving the cab

36. Richtungsschalter to Forward; hold the ZS (train supply) switch at Start until 'ZS Aus' goes out. (Speedbrake lever)
37. Release the doors on the platform side; after 'Schließvorgang eingeleitet' and Zp 9, back to Lock All. (RUD TRIM)
38. **BACK WALL** Parking brake off: with 1.5 bar or more C-pressure press the right (release) button.
39. Führerbremsventil to Running, release the direct brake (keep it on until power on a gradient). (Right throttle)
40. Fahrschalter through Min upwards; or AFB on ('AFB' three times), Fahrschalter to 0 once, set the speed. (Left throttle) (Flap lever)
41. On the move: SiFa reset, PZB Wachsam, Frei, Befehl 40. (Gear lever) (ENG 2) (ENG 1) (Round aux)
42. Leaving: full service, parking brake on (left button), brake key locked, reverser Off and out. (Speedbrake lever) (PARK BRK)

If it will not move

- Reverser stuck: keep PARK BRK up (the reverser key is held only while it is) and the Fahrschalter at Off ('Z-Steller nicht in Null').
- No pantograph or Hauptschalter: the auxiliary reservoir needs 3.5 bar for the pantograph, 7 bar for the HS (LSS 34F05, or drain below 4.8 bar).
- PARK BRK moves both keys. For the brake-key leak tests, lock the brake key in the cab with the mouse so the reverser key stays in.
- AFB will not take: brake on el with the HSM breaker set, train data entered, brake key unlocked, Forward, Fahrschalter back to 0 once.
- SiFa / PZB dead: open their cut-off valves and set LSS 44F01 first; the aux squares only switch the isolation buttons.

Source: BR 101 & Cab-Car Expert Manual, Final Version 3.0, Train Sim Germany

Grey tags: the TCA control for that step · store.steampowered.com/app/4680100

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